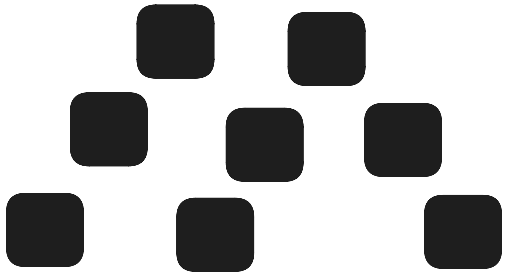
commscube · bananacube · tessa



meilenstein uno

Fabian Fahmüller · Dominik Dethleffsen · Julian Störig · Florian Rößler

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now, the coconut nut is a big, big nut But this delicious nut is not a nut

Problemstellung - Übersicht

Apokalypsen Szenario

- Dezentral
- Minimale Kommunikation gewährleisten
- Unabhänging von jeglicher Infrastruktur
- Einfache Bedienung
- Robust/stoßfest

POC - Proof of Concept

- Kommunikation [1]
- Demo Board [1]



Konzept: [2]



KI Haftungsausschluss: [3]

Hardware - POC

LILYGO LoR32 [1]

- ESP32s3
- LoRa Modul SX1276 (868/915MHz)
- 0.96″ Oled Display
- SMA Female Antenna Connector



LILYGO LoRa32: [1]

Funk - LoRa

Modulation

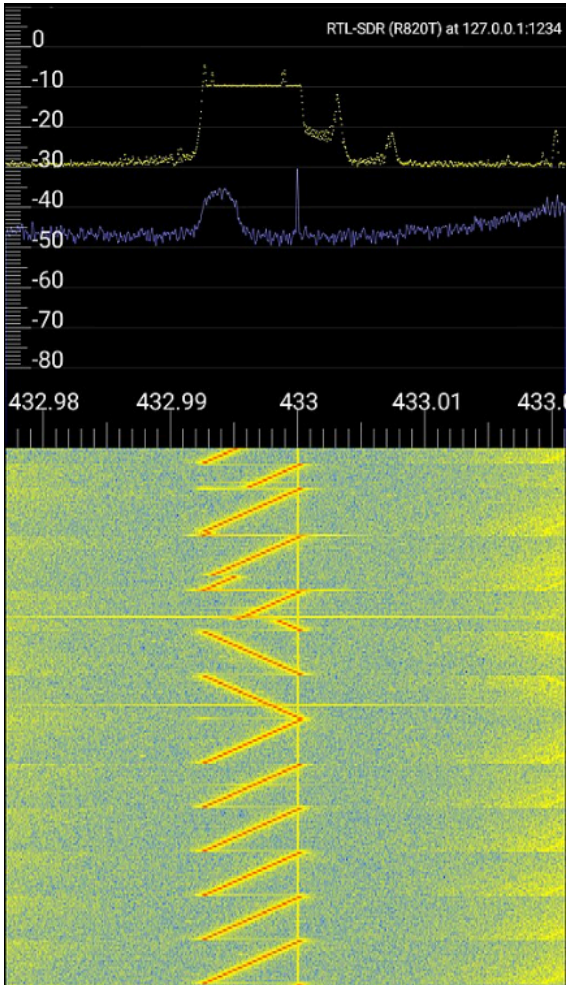
- Chirp Spread Spectrum, Sub-GHz [4]
- SF7–SF12 beim SX1276 [5] wir nutzen SF7
- SF +1: 2x Airtime, +2,5 dB [5]

Eigenschaften

- Praxis 10 km, Rekord 331 km [4], [6]
- Bandbreite 62,5 kHz: im 868-MHz-Band erlaubt, unkritischer Oszillator [5]
- Coding Rate 4/5: Coderate 0,8, reduziert Kanalüberlastung

Regulierung

- Region `EU_868`: 868-MHz-ISM-Band 10 % Duty Cycle, +27 dBm ERP [5]
- 868 MHz: beliebtestes Band in EU [5]



LoRa Modulation: [7]

Protokoll - Meshtastic

Allgemeines

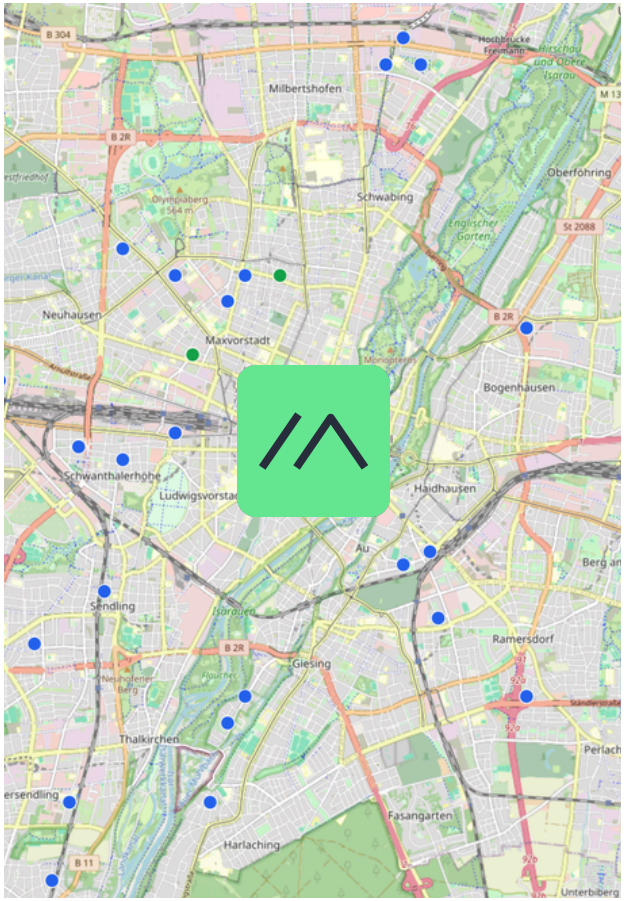
- Open Source, 100 % Community [8]
- Managed Flooding (SNR-basiert) [9]
- Hop Limit 3 (max. 7): Kapazität vor Reichweite [10]
- DMs: Next-Hop-Routing [9]
- Roles: `ROUTER`, `CLIENT`… [9]
- Verschlüsselt: AES, Channel-Key [11]

Commscube Einstellungen

- Preset „Long Fast“ (auch Muc): ca. 156 dB Linkbudget, 1,07 kbit/s [5]
- Sendeleistung `0` (Auto): Rundstrahlantenne < 3 dBi[10]

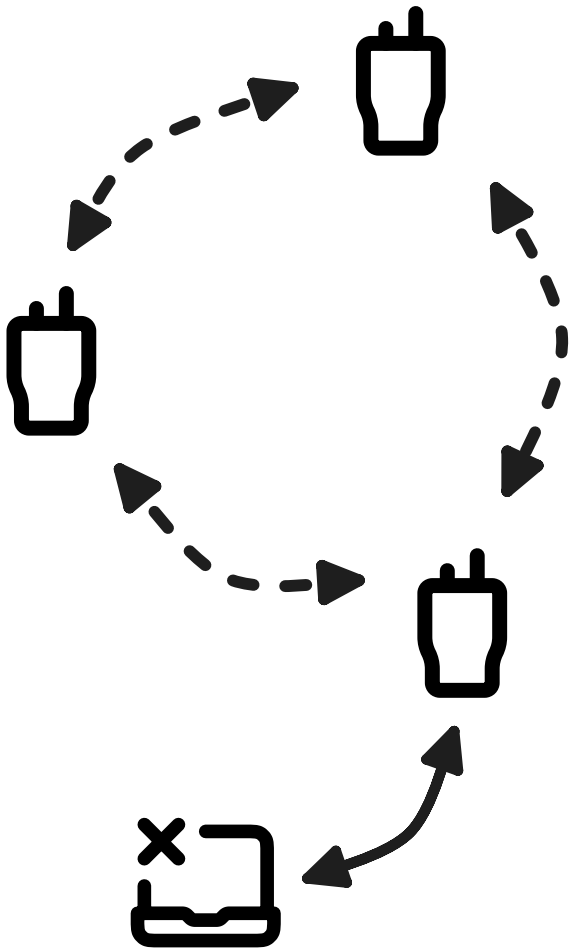
Custom Features

- vordefinierte Nachrichten
- Knopf interface Pin14 Select, Pin4 Send



Meshtastic Logo

Demo Setup



Lokale Node

- Verbunden mit Meshtastic
- Custom Firmware

Remote Node

- Verbunden mit Meshtastic
- Custom Firmware
- Im Roterwürfel

Standard Node

- Verbunden mit Meshtastic
- Standard Firmware
- Per Serial von Laptop gesteuert

Offline Laptop

- locale Website
- Steuert Standard Client per Serial

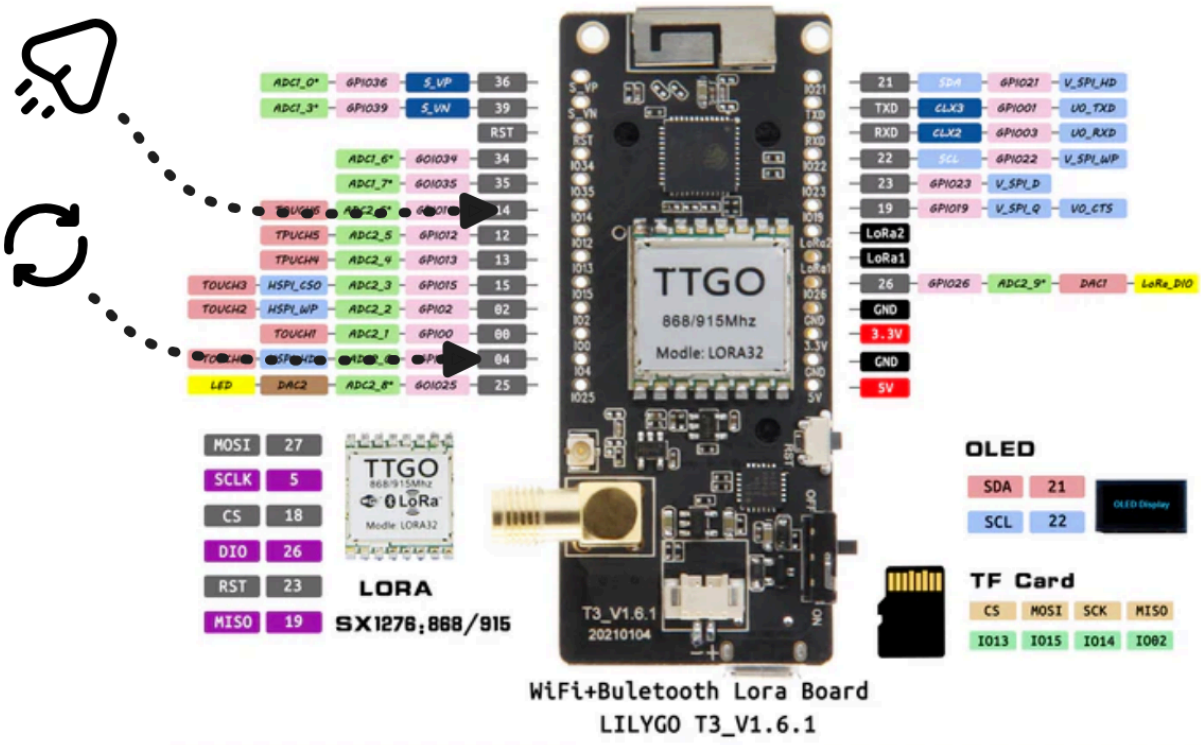
Geführte Demo

Local Demo

- Laptop zu Doku-Cam
- Doku-Cam zu Laptop

Long-range Demo

- Laptop zu “weg”
- “weg” zu Laptop



Pinout Diagram [1]

Interaktive Demo

Toben Sie sich aus Herr Gerstner

Literatur

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[10] Meshtastic, “LoRa Configuration.” Accessed: Nov. 10, 2026. [Online]. Available: https://meshtastic.org/docs/configuration/radio/lora/

[11] Meshtastic, “Meshtastic Encryption.” Accessed: Nov. 10, 2026. [Online]. Available: https://meshtastic.org/docs/overview/encryption/